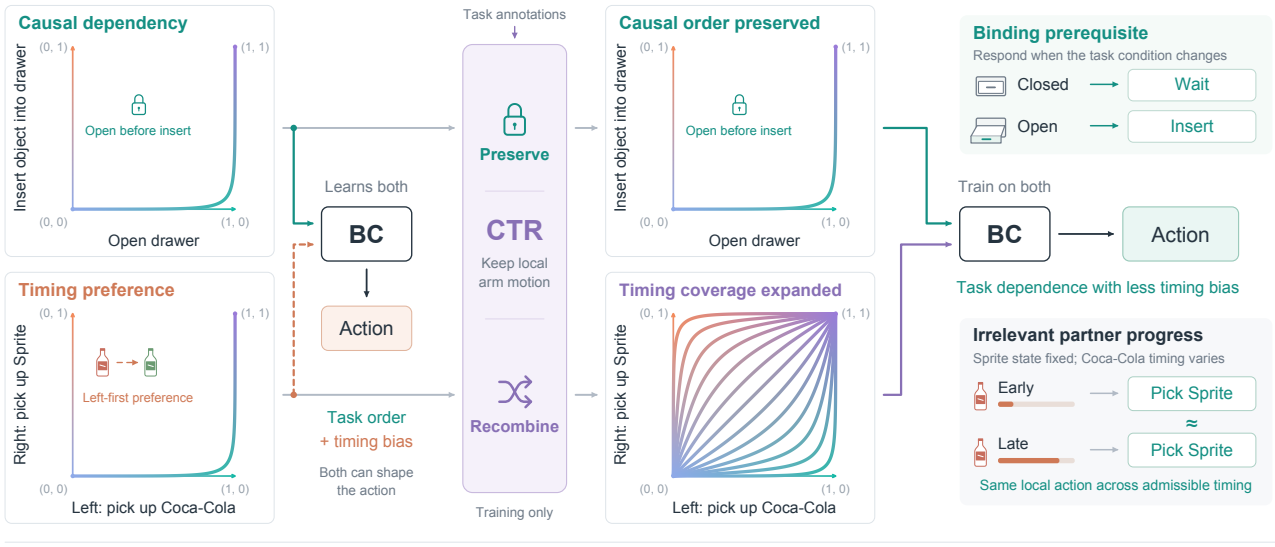

(a) Biased demonstrations
Same observed order, different reasons
Causal dependency
Insert object into drawer
(0, 1)
(1, 1)
Open before insert
(0, 0)
(1, 0)
Open drawer
Timing preference
Right: pick up Sprite
Left-first preference
Left: pick up Coca-Cola
Learns both
BC
Action
Task order
+ timing bias
Both can shape
the action
(b) Constraint-preserving CTR
Counterfactual Temporal Recombination
Task annotations
Preserve
CTR
Keep local
arm motion
Recombine
Training only
Causal order preserved
Timing coverage expanded
(c) Selective action dependence
The same behavior cloning learner
Binding prerequisite
Respond when the task condition changes
Closed
Wait
Open
Insert
Train on both
BC
Action
Task dependence with less timing bias
Irrelevant partner progress
Sprite state fixed; Coca-Cola timing varies
Early
Pick Sprite
≈
Late
Pick Sprite
Same local action across admissible timing
Task-required dependence
Incidental timing bias
Schematic joint-progress coverage; each curve is one relative schedule.